Search for axionlike particle as dark matter candidate with nuclear magnetic resonance

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Abstract

Acknowledgments

Contents

1 Introduction

Interesting questions:

- What is the weakest pseudomagnetic field we can measure? What sensitivity in the exclusion plot does it correspond to?

What is nuclear magnetic resonance? What are axions and axionlike particles?

1.1 Nuclear magnetic resonance

1.1.1 Zeeman splitting and polarization

1.1.2 Spin-projection noise

1.1.3 Relaxations

T_1 relaxation

T_2^* relaxation

Cross relaxation

1.1.4 Chemical shift

1.1.5 Spin-spin coupling

1.1.6 Experimental schemes

Free decay

Spin echo

1.2 Dark matter hypothesis

1.2.1 History

1.2.2 Recent developments

1.3 Axions and axionlike particles

1.3.1 QCD axion

1.3.2 Axionlike particles

1.3.3 Non-gravitational interactions

including *Experiment design for axion research*

1.4 CASPEr project

Idea and method.

CASPEr-E

CASPEr comagnetometers

CASPEr LF, HF

2 Axionlike-particle-induced spin precession

Key questions to be answered:

- How is the interaction between axionlike particles and spins?
- What does the axionlike-particle-induced spin precession signal look like?
- What is the optimal way to detect the signal?

How is the interaction between axionlike particles and spins? ...questions here. By answering these questions in this chapter, we can characterize the signal and determine a good strategy for the experiment.

Part of the content related to the frequency scanning is adapted from the Ref. [1]

2.1 Stochasticity

2.2 Periodic modulations

2.2.1 Daily modulation

2.2.2 Annual modulation

2.3 Spin-precession signal

2.3.1 Relaxations and coherence

2.3.2 Estimation

2.3.3 Numerical simulation

Low frequency: long ALP coherence time, short measurement time

medium frequency

high frequency: short ALP coherence time, short T2star, long T2, long T1

considering the T1 relaxation of hyperpolarized Xenon, we have ... factor of 1/3 worse sensitivity.

2.3.4 Summary

3 Sample and Apparatus

Interesting questions to be answered:

- What are the experimental apparatus?
- What are the uses of them?
- What happens during the experiment?

meta

3.1 Overview

3.2 Magnets

3.3 Coil system

kitchen roll

3.4 Detection chain

mention 1- and 3-coil pickups.

3.5 Data processing

4 Results and analysis

Key questions to be answered:

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-
-

5 Conclusions & Outlook

Key questions to be answered:

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-
-

A Numerical simulation with axionbloch

Key questions to be answered:

-
-
-

???

A.1 Hypothesis test

Key questions to be answered:

- How to set up hypothesis?
-
-

A.2 The Neyman–Pearson Lemma

The *Neyman–Pearson lemma* is a foundational result in statistical hypothesis testing. It establishes the form of the most powerful test for discriminating between two simple hypotheses at a fixed significance level.

A.2.1 Problem Setup

Suppose we observe data x drawn from a probability distribution. We wish to test

$$H_0 : x \sim f_0(x)$$

against

$$H_1 : x \sim f_1(x),$$

where both $f_0(x)$ and $f_1(x)$ are fully specified probability density functions. Such hypotheses are referred to as *simple hypotheses*.

The goal is to construct a statistical test that

- maintains a fixed probability of false alarm (Type I error),
- while maximizing the probability of correctly detecting the alternative hypothesis (power).

A.2.2 Likelihood Ratio

The key quantity is the likelihood ratio

$$\Lambda(x) = \frac{f_1(x)}{f_0(x)}.$$

This ratio quantifies how much more likely the observed data is under the alternative hypothesis than under the null hypothesis.

Large values of $\Lambda(x)$ indicate that the observation is more consistent with H_1 than with H_0.

A.2.3 Statement of the Neyman–Pearson Lemma

For a fixed significance level α, the most powerful test of H_0 against H_1 is given by

$$\Lambda(x) > k,$$

where the threshold k is chosen such that

$$P(\Lambda(x) > k \mid H_0) = \alpha.$$

Equivalently, the optimal decision rule is:

$$\text{Reject } H_0 \quad \text{if} \quad \frac{f_1(x)}{f_0(x)} > k.$$

The lemma therefore establishes that the likelihood-ratio test is optimal among all tests with the same false-positive rate.

A.2.4 Type I Error and Power

The significance level is defined as

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}),$$

which represents the probability of falsely rejecting the null hypothesis.

The power of the test is

$$1 - \beta = P(\text{reject } H_0 \mid H_1 \text{ true}),$$

which measures the probability of correctly detecting the alternative hypothesis.

The Neyman–Pearson lemma guarantees that the likelihood-ratio test maximizes the power for a fixed α.

A.2.5 Gaussian Example

Consider the case

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

with known variance σ^2. We test

$$H_0 : \mu = 0$$

against

$$H_1 : \mu = 1.$$

Under the null hypothesis,

$$f_0(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right),$$

while under the alternative hypothesis,

$$f_1(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-1)^2}{2\sigma^2}\right).$$

The likelihood ratio is therefore

$$\Lambda(x) = \frac{f_1(x)}{f_0(x)} = \exp\left[-\frac{(x-1)^2}{2\sigma^2} + \frac{x^2}{2\sigma^2}\right].$$

Expanding the exponent,

$$(x-1)^2 = x^2 - 2x + 1,$$

so that

$$\log \Lambda(x) = \frac{x}{\sigma^2} - \frac{1}{2\sigma^2}.$$

Hence, rejecting for large $\Lambda(x)$ is equivalent to rejecting for sufficiently large x.

Bibliography

[1] Yuzhe Zhang, Deniz Aybas Tumturk, Hendrik Bekker, Dmitry Budker, Derek F. Jackson Kimball, Alexander O. Sushkov, and Arne Wickenbrock. **Frequency-Scanning Considerations in Axionlike Dark Matter Spin-Precession Experiments**. *Annalen der Physik* 536:1 (2024), 2300223. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/andp.202300223 (see page 5).